Yuan-Yen Peng

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Research Interests

- ML for LHC physics with physics-constrained, attention, and flow-based architectures
- Inference and event reconstruction for under-constrained and compressed new physics
- Searches for physics beyond the SM, including SUSY and dark-matter–motivated models
- Higgs and EW processes as precision control and validation channels for new physics

Education

MSc in Physics, National Tsing Hua University — Sept. 2023 – June 2025
GPA: 4.26 / 4.30 — Hsinchu, Taiwan
Thesis: *Quantum Entanglement Measurements in the $H \to WW^* \to \ell\nu\ell\nu$ Decay Channel with the ATLAS Detector*

BSc in Physics, National Tsing Hua University — Aug. 2020 – June 2023
GPA: 3.98 / 4.30 (Last 60 credits: 4.12) — Hsinchu, Taiwan

Experimental High Energy Physics Summer School — June 2022
Taiwan Instrumentation and Detector Consortium — Taipei, Taiwan

Tsing Hua Interdisciplinary Program — Sept. 2019 – July 2020
National Tsing Hua University — Hsinchu, Taiwan

Academic Experience

R&D Substitute Military Service — July. 2025 – Jan. 2027
ATLAS Group in Natl. Tsing Hua Univ. — Hsinchu, Taiwan

- Developed **attention-based, physics-constrained, invertible neural networks** for $H \to WW^* \to \ell\nu\ell\nu$ under-constrained problems.
- Measured the spin-spin correlations of WW^* from different polarizations.
- Co-authored **ATLAS internal note (ANA-HIGP-2024-27-INT1)** on ML-based reconstructions and interpretation of W polarizations.
- Participated in the HWW^* Run3 coupling analysis within the ATLAS group.
- Co-developed frontend and backend of High Granularity Timing Detector production database in the ATLAS.
- Built and maintained the high-performance AI and local CernVM-FS caching workstation with containerized environments.
- Co-mentored students on ATLAS analysis and ML tools.

Research Assistant — Sept. 2022 – June 2025

- Measured quantum entanglement in $H \to WW^* \to \ell\nu\ell\nu$ at the ATLAS experiment.
- Delivered several talks in ATLAS meetings.

Research Student — Feb. 2021 – Sept. 2022

- Early-stage ATLAS analysis experience and hands-on HEP software frameworks.

Teaching Assistant Feb. 2024 – Jan. 2025
Dept. of Physics in Natl. Tsing Hua Univ. Hsinchu, Taiwan

- Graduate-level Quantum Mechanics I
- Thermal and Statistical Physics I
- General Physics B II

ATLAS Workshops and Collaboration Activities

ATLAS NRML Workshop Apr. 2025
at Nikhef Amsterdam, Netherlands

- [Oral presentation]: *Reconstructions of W and ν with Machine Learning*

ATLAS Collaboration Week Oct. 2024
at CERN Genève, Switzerland

- [Oral presentation]: *Reconstruction of W Boson Rest Frames in $H \to WW^* \to \ell\nu\ell\nu$*

Annual Meeting of the Physical Society of Taiwan Jan. 2025
at Natl. Sun Yat-sen Univ. Kaohsiung, Taiwan

- [Poster presentation]: *Quantum Entanglement in $H \to WW^* \to \ell\nu\ell\nu$*

ATLAS HWW Workshop Mar. 2023
at Uni Freiburg Freiburg, Germany

- [Contributed talk]: Oral presented in pre-workshop focus session.

HWW Quantum Entanglement Kick-off Meeting Apr. 2024
via Virtual Meeting Online

- NTHU group contributor.

ATLAS Higgs Workshop Oct. 2022
at UTokyo Tokyo, Japan

- Plenary session contributor.

Honors and Scholarships

Phi Tau Phi Scholastic Honor Society Honorary Member	June 2025
College of Science Elite Student Award, NTHU	April 2025
Maxwell Scholarship, Dept. of Physics, NTHU (merit-based)	May 2024
Ms. Ru-Bao Huang Li Physics Memorial Scholarship, NTHU	Apr. 2024
Mr. Hong-Lin Wu Memorial Scholarship (academic merit)	June 2021
Outstanding Academic Achievement Award, NTHU ($\times$ 3)	2020 – 2023

Technical Skills

ML: PyTorch/Lightning, FrEIA, ONNX, TensorFlow/Keras

HEP: ROOT, FastFrames (HWW), uproot, mplhep

Database & Web: Django (working knowledge), Vue.js (working knowledge)

Programming: Python, C++

Others: Git, Docker/Podman, CernVM-FS, UNIX, OpenCode, Ollama

Markup tools: LaTeX, Markdown, conventional office suites

Languages: English (TOEFL iBT 101), Mandarin Chinese (native)